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Sri Lanka Institute of Information Technology

B.Sc.(Hons) in Computer Science/ Computer Systems Engineering

Final Examination

Year 2, Semester 1 (2024)

IE2024 – Probability and Statistics

Duration: 2 Hours

October, 2024

Instructions to Candidates:

- This paper is preceded by a 10-minutes reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions. Answer all questions in the booklet given.
- The total mark for the paper is 100.
- Nonprogrammable calculators are allowed to be used.
- Other electronic devices capable of storing and retrieving text, including mobile phones are not allowed.
- This paper contains 5 pages, including the cover page.

Question 1 (25 Marks)

Sampling is the process of selecting a subset of individuals or data points from a larger population to represent that population. It helps researchers make inferences about the whole group without studying every individual, saving time and resources. Once a sample is selected, descriptive statistics are used to summarize and describe the main features of the data.

a) Compare and contrast Probability Sampling and Non-Probability Sampling. (3 Marks)

b) Compare and contrast the Stratified Sampling and Cluster Sampling. (3 Marks)

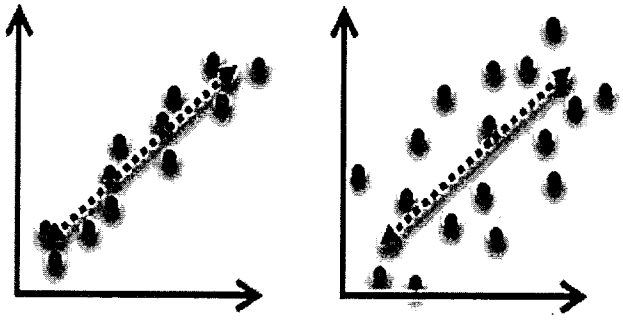
c) Show that if a data distribution has no outliers, the data points must be in the range $[a, b]$ such that,

$$(Q_3 - Q_1) = \frac{(b-a)}{4}$$

where Q_1 and Q_3 are the first and third quartiles in the data distribution. (10 Marks)

d) Critically analyze the statement "Correlation does not imply the causation". (4 Marks)

e) Two scatter plots are given below, and each represents a relationship between two numerical variables. Analyze the relationships of both charts briefly. (5 Marks)



Question 2 (25 Marks)

Probability theory is a branch of mathematics that deals with quantifying uncertainty and analyzing random phenomena. It provides a framework for modeling and understanding uncertain events and making predictions based on data and probability distributions.

a) Suppose that P(A) = 0.4, P(B) = 0.3 and P(A∪B) = 0.58. Are A and B independent? (3 Marks)

b) Let S be a sample space and A and B be two events such that P(A) = 0.8 and P(B) = 0.9. Determine whether A and B are mutually exclusive or not. (4 Marks)

c) The Bernoulli distribution models the probability of a single binary outcome (success or failure) with a fixed probability of success.

$$P(X = x) = \begin{cases} p^x(1-p)^{1-x} & for\ x = 0,1 \\ 0 & Otherwise \end{cases}$$

When p is the empirical probability of success. Show that the expected value of the Bernoulli distribution (E(X)) is **p**. (6 Marks)

d) Suppose a researcher is conducting an experiment to study the number of times rats press a lever to receive food. Let's assume that the rate at which the rats press the lever is 4 times per minute on average.

i. Find the probability that a rat presses the lever exactly 10 times during a 4-minute observation period. (4 Marks)

ii. Suppose that laboratory has a population of rats that are known to have a lifespan of 3 years, on average. Assuming that the lifespans of rats are exponentially distributed. What is the probability that a randomly selected rat will live exactly 2 years? (4 Marks)

iii. Write an integration expression to the probability that a randomly selected rat will live between 2 to 4 years. (No need to solve the integration.) (4 Marks)

Question 3 (25 Marks)

Statistical inference involves using probability theory and statistical methods to make conclusions about a population based on information from a sample. It helps us estimate parameters, test hypotheses, and make predictions about future outcomes. However, it requires assumptions and may involve sources of uncertainty.

a) Show that the sample mean $\bar{X}$ is an unbiased estimator for the population mean μ when

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

where n is the sample size. (5 Marks)

b) Compare and contrast the Type I error and Type II error in Hypothesis Testing. (4 Marks)

c) From a class 4 male students and 3 female students were drawn and their mathematics marks were checked. The average mark in the male sample is 72 and the variance is 132. For the female sample the average is 60 and the variance is 100. Assuming the male and female groups are normally distributed and have equal variances, find 95% confidence interval for the difference between the mean of the mathematics marks for male and female. ($t_{2.5\%,5}$=2.57) (6 Marks)

d) The same test was given to another group of 100 males and 144 females. The mean score for males was 78 and the mean score for the females was 74. Assuming a common population standard deviation of 3. Use hypothesis testing to test whether the male's performance in the test was better than that of the females.

   i. Write the Null & Alternative hypotheses for the above scenario. (2 Marks)

   ii. Test at 5% level of significance whether the male's performance in the test was better than that of the females. Assume that the scores are normally distributed. ($z_{95\%} = 1.645$) (8 Marks)

Question 4 (25 Marks)

A variable's value can be predicted using linear regression analysis based on the value of another variable. The dependent variable is the one you want to be able to forecast. The independent variable is the one you're using to make a prediction about the value of the other variable. Consider the following dataset. Here are the first few rows of the dataset. Here the objective is to predict the GPA using other independent variables through a Linear Regression Model. (The first few rows of the dataset have been given just to get an idea about the variables. Don't use this data to calculations)

ID	Gender	Age	Extra_Curricular	Study_Hours	Annual_Income	Distance_From_Home	GPA
8867	Male	22	Societies	10	1318792	154	1.76
1316	Female	24	Societies	9	874657	181	3.88
9443	Male	22	Societies	15	1098331	94	1.73
1829	Female	25	No	9	1680585	8	1.29
1989	Male	26	No	7	1361560	122	3.19
7297	Female	25	Societies	5	887824	107	2.42
7447	Male	23	Societies	10	1957325	22	3.05
2018	Male	21	Societies	7	1602856	130	2.14
1136	Female	23	Societies	8	429437	124	2.07
7702	Male	26	Sports	6	536744	81	1.85

a) Mention two assumptions in Linear Regression and briefly explain. (4 Marks)

b) Gender and the Extra Curricular variables are categorical independent variables. Gender variable has two categories, and Extra Curricular variable has three categories as mentioned in the above snapshot. Explain how these two variables can be converted to dummy variables? (6 Marks)

c) Assume for the above linear regression model,

$$\text{Total Sum of Squares (TSS)} = 1000$$

$$\text{Sum of Squares of Error (SSE)} = 140$$

Find the Coefficient of Determination for this model using the above values. (5 Marks)

d) Analyze the goodness of fit of the above model using the Coefficient of Determination value, calculated in part (c). (4 Marks)

e) Assume the estimated simple linear regression model for predicting the GPA with the Study Hours is as below.

$$\widehat{GPA} = 0.1 + 0.35 \times \widehat{Study_Hours}$$

a. Interpret the intercept (0.1) of the model (3 Marks)

b. Interpret the slope (0.35) of the model (3 Marks)

----------End of the Question Paper ----------